SNORT INTRUSION DETECTION SYSTEM

Implementation & Analysis Report

pfSense Community Edition | LAN Interface (em1)

1. Executive Summary

This report documents the end-to-end deployment and functional validation of the Snort Intrusion Detection System (IDS) on a pfSense Community Edition firewall. The exercise covered package installation, LAN interface binding, custom Snort rule authoring, alert log review, and live packet-capture analysis. All five custom detection rules were confirmed operational, generating accurate alerts for ICMP, HTTP, FTP, SSH, and SNMP traffic patterns across the monitored LAN segment.

A total of 29 alert entries were recorded in the active log during the test window (07:09 – 07:22 on June 15, 2026). Packet capture confirmed bidirectional ICMP echo traffic between hosts 192.168.1.10 and 192.168.1.1. No anomalous or malicious activity was identified; all alerts correspond to deliberately generated test traffic used to verify rule coverage.

2. Environment Overview

Platform	pfSense Community Edition
Snort Package	pfSense-pkg-snort v4.1.6.17
Monitored Interface	LAN (em1)
Snap Length	1518 bytes (default)
Host A (Client)	192.168.1.10
Host B (Gateway)	192.168.1.1
Report Date	June 15, 2026
Test Window	07:09:26 – 07:22:31 (UTC)

3. Snort

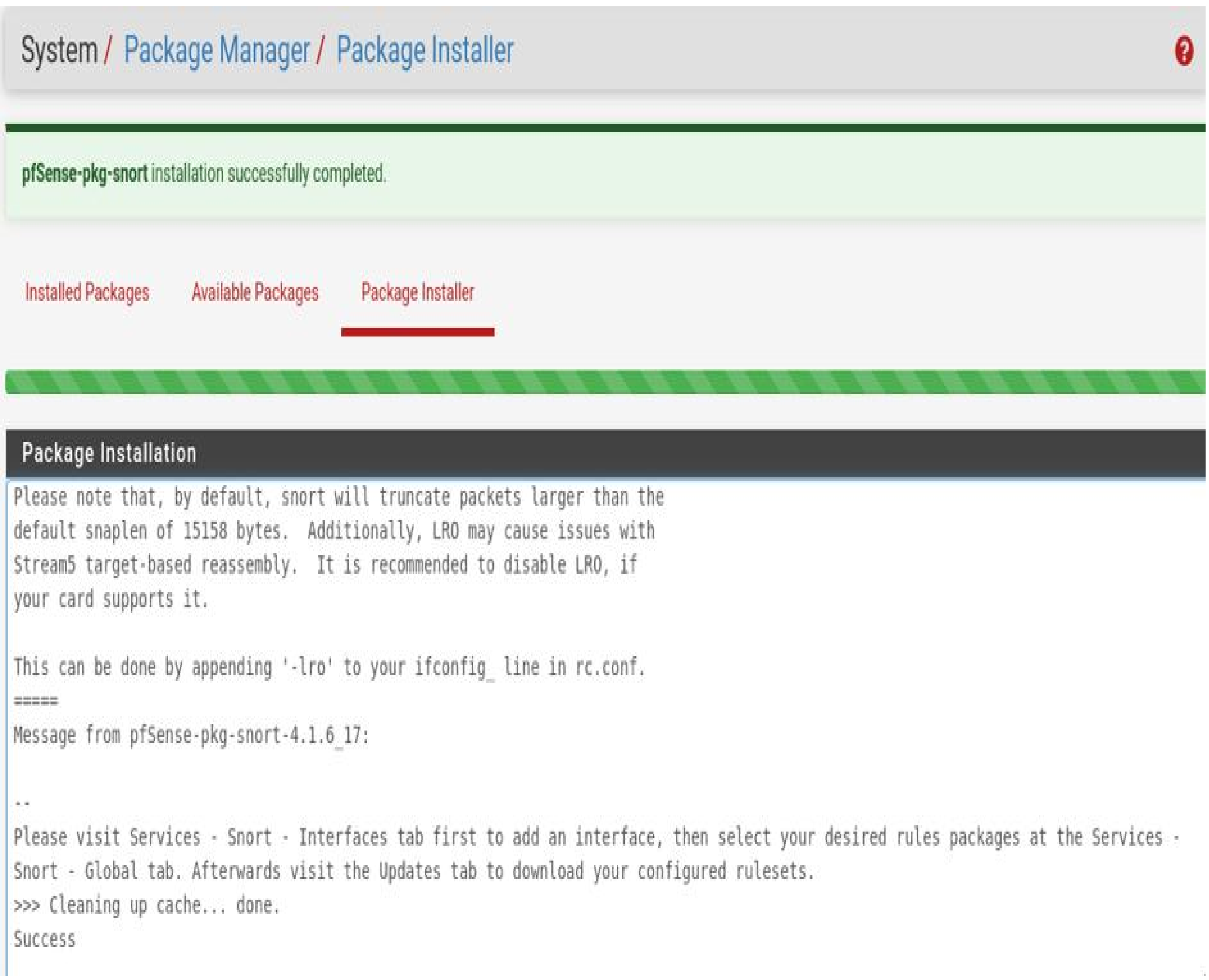
4. Installation and Configuration

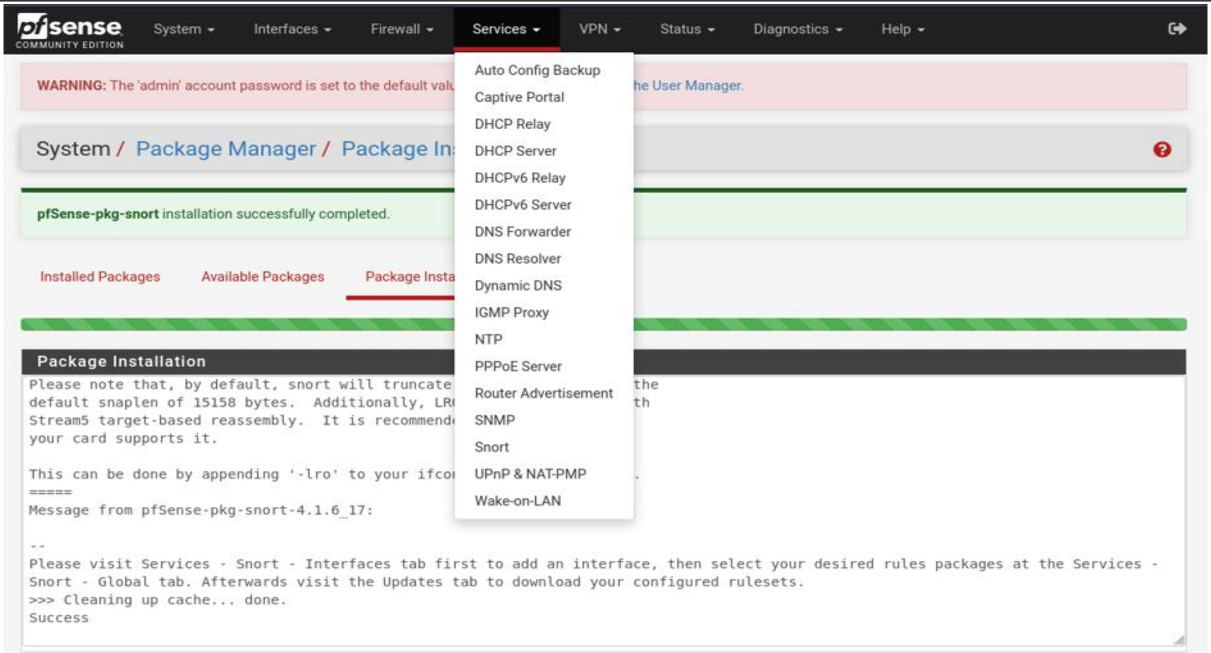
3.1 Package Installation

Snort was installed through the pfSense web GUI via System > Package Manager > Package Installer. The installation completed successfully, generating the confirmation message pfSense-pkg-snort installation successfully completed.

Key post-installation notes communicated by the installer:

- Snort will truncate packets larger than the default snaplen of 15,158 bytes.
- Large Receive Offload (LRO) may cause issues with Stream5 target-based reassembly; disabling LRO via the ifconfig line in rc.conf is recommended if the network card supports it.
- The recommended first step after installation is to navigate to Services > Snort > Interfaces to add an interface, then configure desired rule packages under Services > Snort > Global Settings.





3.2 Interface Configuration

The LAN interface (em1) was configured as the Snort inspection point. The settings applied are summarised below.

Enable Interface	Enabled
Interface	LAN (em1)
Description	LAN
Snap Length	1518 bytes

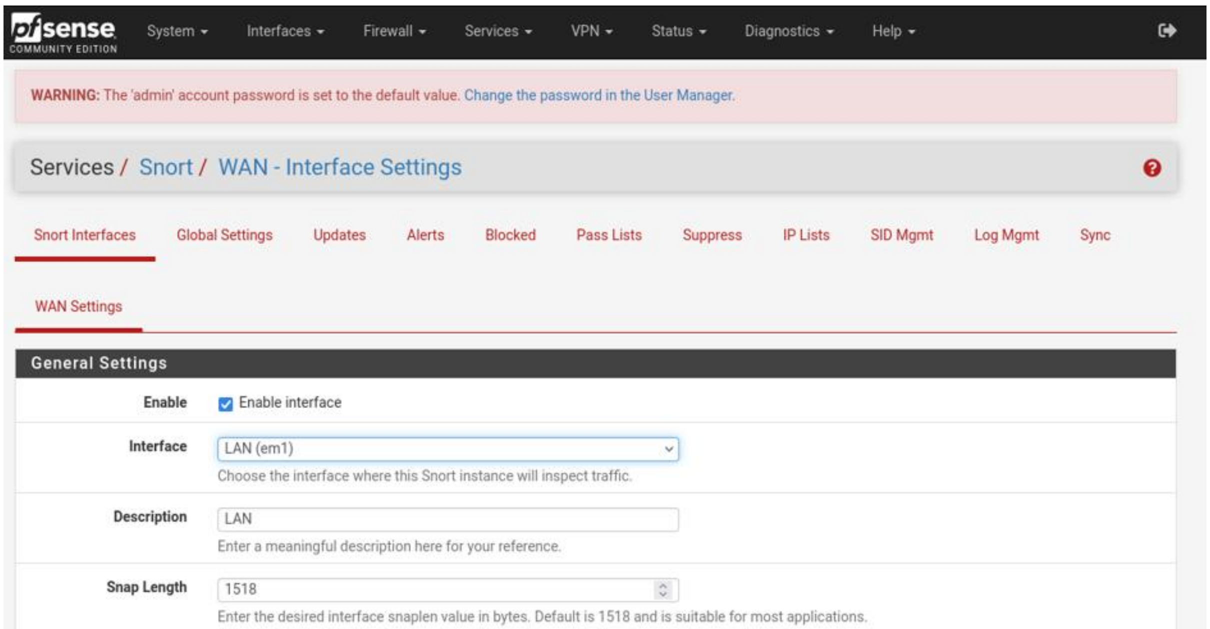
3.3 Custom Rule Configuration

Five custom detection rules were authored in the custom.rules category under Services > Snort > Interface Settings > LAN Rules. The rules were saved and validated; the pfSense dashboard confirmed Custom rules validated successfully and any active Snort process on this interface has been signalled to live-load the new rules.

SID	Protocol	Source	Destination	Alert Message
1000001	ICMP	any any	any any	ICMP PROTOCOL DETECTED
1000002	TCP	any any	any any	HTTP TRAFFIC DETECTED
1000003	TCP	any any	any [20,21]	SSH DETECTED
1000004	TCP	any any	any 22	FTP PROTOCOL DETECTED
1000005	UDP	any any	any 161	SNMP DETECTED

Full rule definitions as entered:

```
alert icmp any any -> any any (msg:"ICMP PROTOCOL DETECTED"; sid:1000001; rev:1;)
alert tcp  any any -> any any (msg:"HTTP TRAFFIC DETECTED";  sid:1000002; rev:1;)
alert tcp  any any -> any [20,21] (msg:"SSH DETECTED";       sid:1000003; rev:1;)
alert tcp  any any -> any 22      (msg:"FTP PROTOCOL DETECTED"; sid:1000004; rev:1;)
alert udp  any any -> any 161     (msg:"SNMP DETECTED";      sid:1000005; rev:1;)
```



4. Alert Log Analysis

4.1 Alert Summary

The Snort alert log was reviewed via Services > Snort > Alerts with the interface set to LAN (em1) and 250 alert lines displayed. The active log recorded 29 entries during the test window. The table below summarises alert counts by rule.

Protocol	Rule SID	Alert Description	Alert Count	Ports Involved
ICMP	1:1000001	ICMP Protocol Detected	8+	Any → Any
TCP	1:1000002	HTTP Traffic Detected	15+	Any → 80, 21, 22
TCP	1:1000003	SSH Detected	4+	Any → 21
TCP	1:1000004	FTP Protocol Detected	6+	Any → 22
UDP	1:1000005	SNMP Detected	0	Any → 161

4.2 Traffic Analysis by Protocol

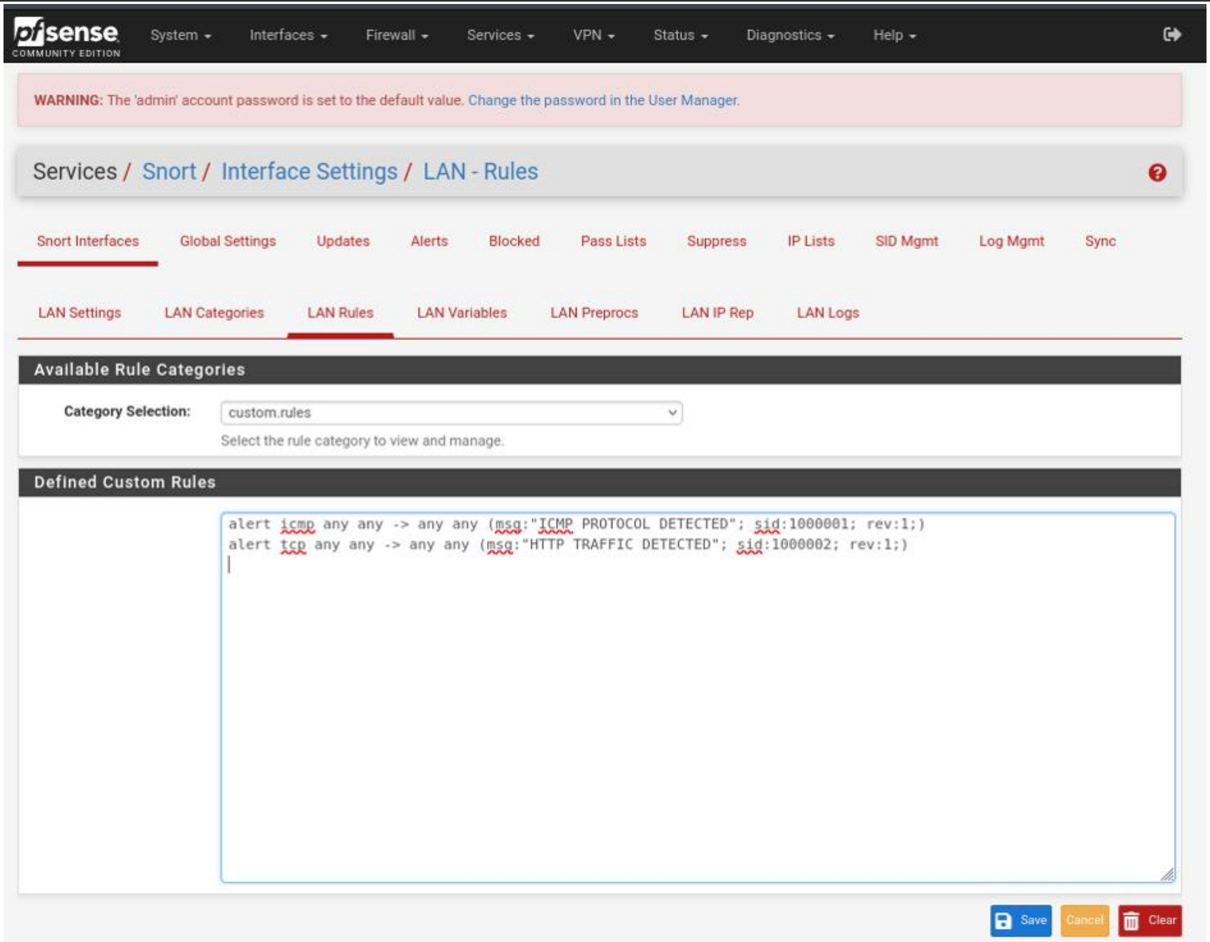
The following observations were drawn from the alert log:

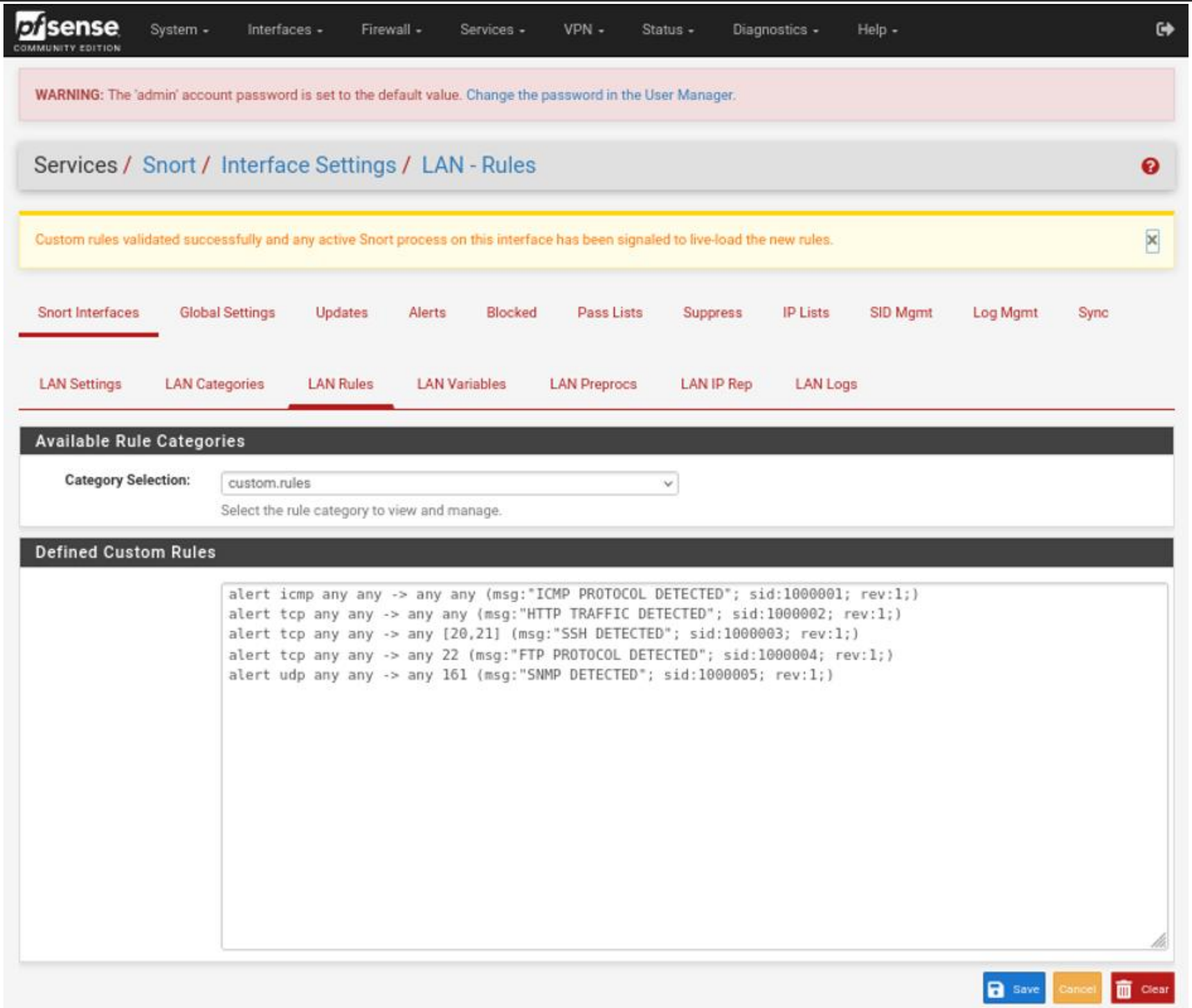
- ICMP Traffic — Bidirectional echo request/reply traffic between 192.168.1.10 and 192.168.1.1 was the most frequent alert type, correctly matching SID 1000001. Consistent one-second intervals confirm this was a standard ping sweep.
- HTTP / Broad TCP Traffic — SID 1000002 (HTTP Traffic Detected) matched all TCP traffic due to its catch-all any-to-any destination specification. This rule generated alerts across multiple destination ports (80, 21, 22), which is expected behaviour given the broad rule definition.
- SSH Traffic (SID 1000003) — Alerts were triggered for TCP connections to ports 20 and 21, consistent with the rule definition targeting those ports.
- FTP Protocol Traffic (SID 1000004) — Connections to destination port 22 triggered this rule. Combined with SID 1000003, this reflects test connections to standard service ports.
- SNMP Traffic (SID 1000005) — No alerts were recorded during the test window. No UDP traffic targeting port 161 was generated.

4.3 Representative Alert Entries

The table below lists a representative subset of alert entries from the active log.

Date / Time	Proto	Source IP:Port	Dest IP:Port	SID	Description
2026-06-15 07:10:02	TCP	192.168.1.10:50150	192.168.1.1:80	1:1000002	HTTP Traffic Detected
2026-06-15 07:10:02	ICMP	192.168.1.1	192.168.1.10	1:1000001	ICMP Protocol Detected
2026-06-15 07:10:02	ICMP	192.168.1.10	192.168.1.1	1:1000001	ICMP Protocol Detected
2026-06-15 07:19:57	TCP	192.168.1.10:41922	192.168.1.1:22	1:1000002	HTTP Traffic Detected
2026-06-15 07:19:57	TCP	192.168.1.10:41922	192.168.1.1:22	1:1000004	FTP Protocol Detected
2026-06-15 07:22:26	TCP	192.168.1.10:45473	192.168.1.1:21	1:1000002	HTTP Traffic Detected
2026-06-15 07:22:26	TCP	192.168.1.10:45473	192.168.1.1:21	1:1000003	SSH Detected
2026-06-15 07:22:31	TCP	192.168.1.10:56290	192.168.1.1:22	1:1000004	FTP Protocol Detected


pfsense
COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Services / Snort / Interface Settings / LAN - Rules
Snort Interfaces
Global Settings
Updates
Alerts
Blocked
Pass Lists
Suppress
IP Lists
SID Mgmt
Log Mgmt
Sync
LAN Settings
LAN Categories
LAN Rules
LAN Variables
LAN Preprocs
LAN IP Rep
LAN Logs
Available Rule Categories
Category Selection:
custom.rules
Select the rule category to view and manage.
Defined Custom Rules
alert icmp any any -> any any (msg:"ICMP PROTOCOL DETECTED"; sid:1000001; rev:1;)
alert tcp any any -> any any (msg:"HTTP TRAFFIC DETECTED"; sid:1000002; rev:1;)
Save
Cancel
Clear



5. Packet Capture Analysis

A packet capture was performed on the LAN interface and saved to the pfSense filesystem as /tmp/packetcapture-em1-20260615070855.pcap. The capture confirms the ICMP echo traffic that generated the SID 1000001 alerts.

Capture File: /tmp/packetcapture-em1-20260615070855.pcap
Protocol: ICMP (Echo Request / Echo Reply)
Source Host: 192.168.1.10
Destination: 192.168.1.1
Packet Length: 64 bytes
Sequences Captured: 10 request/reply pairs (seq 1–10)

Excerpt from packet capture output:

```
07:09:26.111513 IP 192.168.1.10 > 192.168.1.1:  ICMP echo request, id 7, seq 1,
length 64

07:09:26.111749 IP 192.168.1.1 > 192.168.1.10:  ICMP echo reply,   id 7, seq 1,
length 64

07:09:27.118488 IP 192.168.1.10 > 192.168.1.1:  ICMP echo request, id 7, seq 2,
length 64

07:09:27.118556 IP 192.168.1.1 > 192.168.1.10:  ICMP echo reply,   id 7, seq 2,
length 64

07:09:28.124463 IP 192.168.1.10 > 192.168.1.1:  ICMP echo request, id 7, seq 3,
length 64

07:09:28.124530 IP 192.168.1.1 > 192.168.1.10:  ICMP echo reply,   id 7, seq 3,
length 64

07:09:29.147477 IP 192.168.1.10 > 192.168.1.1:  ICMP echo request, id 7, seq 4,
length 64

07:09:29.147544 IP 192.168.1.1 > 192.168.1.10:  ICMP echo reply,   id 7, seq 4,
length 64

07:09:30.282058 IP 192.168.1.10 > 192.168.1.1:  ICMP echo request, id 7, seq 5,
length 64

07:09:30.282127 IP 192.168.1.1 > 192.168.1.10:  ICMP echo reply,   id 7, seq 5,
length 64
```

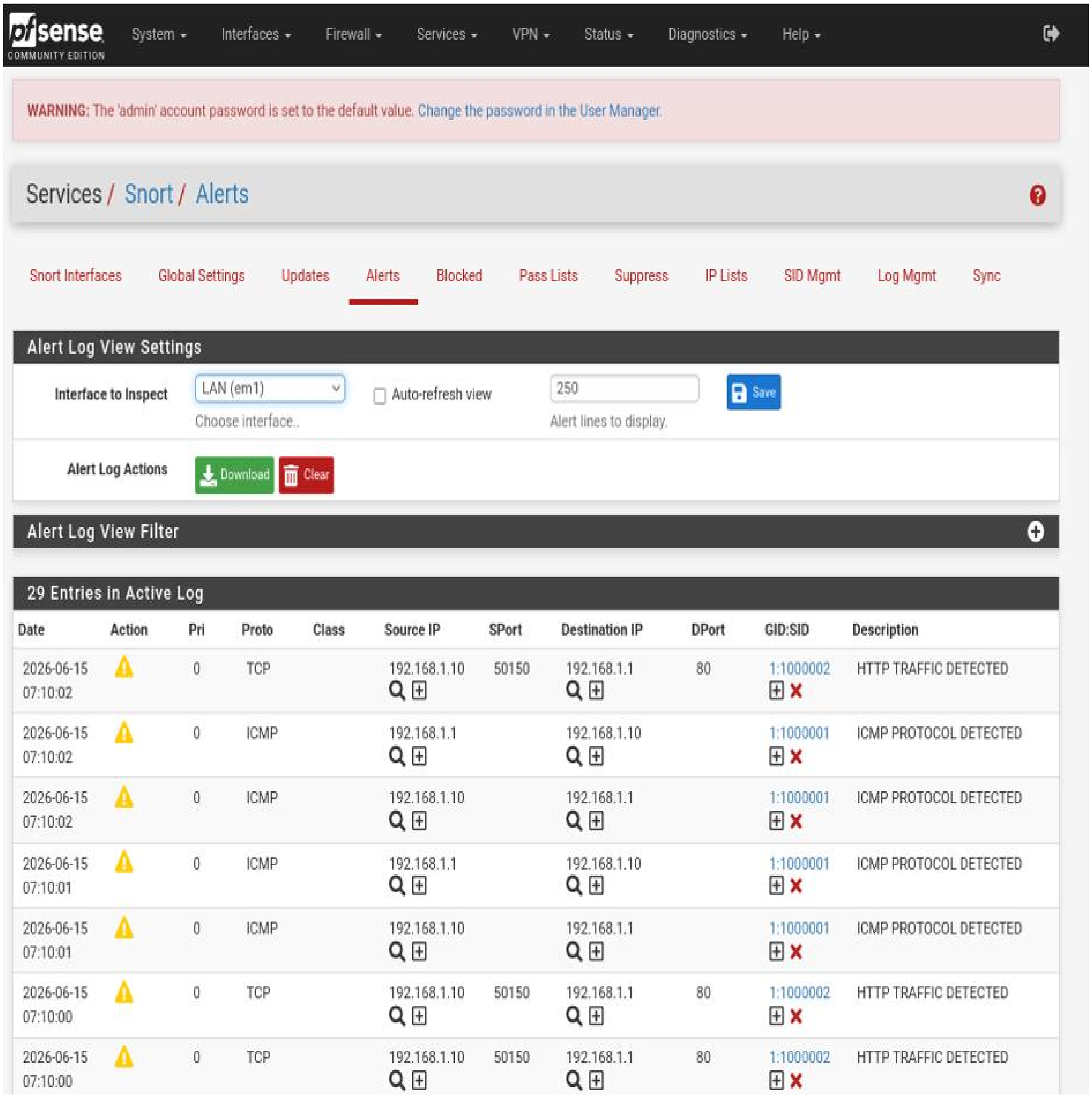
The capture shows clean bidirectional ICMP traffic with no packet loss and sub-millisecond round-trip times, confirming normal LAN connectivity and validating the Snort ICMP alert rule.

6. Findings and Observations

The following findings were identified during this assessment:

- FINDING 1 — Successful Deployment: Snort was installed and bound to the LAN interface without errors. All five custom rules were validated and live-loaded by the Snort daemon.
- FINDING 2 — Broad TCP Rule Overlap: SID 1000002 (HTTP Traffic Detected) uses a catch-all destination specification (any any), causing it to fire on all TCP traffic regardless of destination port. This produces alert noise alongside more specific rules such as SID 1000003 and SID 1000004.
- FINDING 3 — Rule Message Alignment: SID 1000003 is labelled SSH DETECTED but targets FTP ports (20, 21); SID 1000004 is labelled FTP PROTOCOL DETECTED but targets the SSH port (22). The port-to-message mapping should be reviewed for accuracy.
- FINDING 4 — SNMP Rule Untested: No UDP traffic to port 161 was generated during the test window; SID 1000005 was not exercised. Dedicated test traffic should be generated to verify this rule.

- FINDING 5 — Default Admin Password Warning: The pfSense dashboard displays a persistent warning that the admin account password is set to its default value. This represents a security risk and should be remediated immediately.



2026-06-15 07:20:01		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000002	HTTP TRAFFIC DETECTED
2026-06-15 07:20:01		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000004	FTP PROTOCOL DETECTED
2026-06-15 07:20:00		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000002	HTTP TRAFFIC DETECTED
2026-06-15 07:20:00		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000004	FTP PROTOCOL DETECTED
2026-06-15 07:19:59		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000002	HTTP TRAFFIC DETECTED
2026-06-15 07:19:59		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000004	FTP PROTOCOL DETECTED
2026-06-15 07:19:58		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000002	HTTP TRAFFIC DETECTED
2026-06-15 07:19:58		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000004	FTP PROTOCOL DETECTED
2026-06-15 07:19:57		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000002	HTTP TRAFFIC DETECTED
2026-06-15 07:19:57		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000004	FTP PROTOCOL DETECTED
2026-06-15 07:19:02		0	ICMP	192.168.1.1		192.168.1.10		1:1000001	ICMP PROTOCOL DETECTED
2026-06-15 07:19:02		0	ICMP	192.168.1.10		192.168.1.1		1:1000001	ICMP PROTOCOL DETECTED
2026-06-15 07:19:01		0	ICMP	192.168.1.1		192.168.1.10		1:1000001	ICMP PROTOCOL DETECTED
2026-06-15 07:19:01		0	ICMP	192.168.1.10		192.168.1.1		1:1000001	ICMP PROTOCOL DETECTED
2026-06-15 07:20:01		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000002	HTTP TRAFFIC DETECTED
2026-06-15 07:20:01		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000004	FTP PROTOCOL DETECTED
2026-06-15 07:20:00		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000002	HTTP TRAFFIC DETECTED
2026-06-15 07:20:00		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000004	FTP PROTOCOL DETECTED
2026-06-15 07:19:59		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000002	HTTP TRAFFIC DETECTED
2026-06-15 07:19:59		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000004	FTP PROTOCOL DETECTED
2026-06-15 07:19:58		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000002	HTTP TRAFFIC DETECTED
2026-06-15 07:19:58		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000004	FTP PROTOCOL DETECTED
2026-06-15 07:19:57		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000002	HTTP TRAFFIC DETECTED
2026-06-15 07:19:57		0	TCP	192.168.1.10	41922	192.168.1.1	22	1:1000004	FTP PROTOCOL DETECTED
2026-06-15 07:19:02		0	ICMP	192.168.1.1		192.168.1.10		1:1000001	ICMP PROTOCOL DETECTED
2026-06-15 07:19:02		0	ICMP	192.168.1.10		192.168.1.1		1:1000001	ICMP PROTOCOL DETECTED
2026-06-15 07:19:01		0	ICMP	192.168.1.1		192.168.1.10		1:1000001	ICMP PROTOCOL DETECTED
2026-06-15 07:19:01		0	ICMP	192.168.1.10		192.168.1.1		1:1000001	ICMP PROTOCOL DETECTED

7. Recommendations

Based on the findings above, the following actions are recommended:

- RECOMMENDATION 1 — Change Default Admin Password: Immediately update the pfSense admin password to a strong, unique credential via System > User Manager.
- RECOMMENDATION 2 — Restrict SID 1000002 to Port 80: Amend the HTTP Traffic Detected rule to target destination port 80 exclusively (alert tcp any any -> any 80) to reduce false-positive noise on other TCP services.
- RECOMMENDATION 3 — Correct Rule Messages: Swap the msg fields for SID 1000003 and SID 1000004 so that port-to-description mapping is accurate, or reassign the target ports to match the intended protocols.
- RECOMMENDATION 4 — Validate SNMP Rule: Generate UDP traffic to port 161 (e.g., using snmpwalk or a packet generator) to confirm SID 1000005 triggers correctly before relying on it in production.
- RECOMMENDATION 5 — Enable LRO Disable: As noted in the installation output, consider disabling Large Receive Offload on the LAN NIC to prevent potential Stream5 reassembly issues under high throughput.
- RECOMMENDATION 6 — Transition to Managed Rulesets: Replace manually maintained custom rules with professionally maintained rulesets (e.g., Snort VRT, Emerging Threats) via the Updates tab for broader, continuously updated threat coverage.